

# Gas Absorption

Final Lab Report

Unit Operations Lab 6

April 20, 2026

Group 3

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## 1. Executive Summary

Using an Armfield Gas Absorption Column (UOP7), the objectives of this lab are to investigate the effects of gas and water flowrates on the column's pressure drop, as well as to determine the overall mass transfer coefficient of in both an open and closed setup. A summary of the overall mass transfer coefficient under different operating conditions is provided below

*Table 1: Summary of CO<sub>2</sub> Mass Transfer Coefficient*

| Operating Condition         | Mass Transfer Coefficient |
|-----------------------------|---------------------------|
| Closed-loop Absorption (2a) | 0.3674                    |
| Desorption (2b)             | 0.861                     |
| Open-Loop Absorption (3)    | 0.00535                   |

Overall, Table 1 showed that the closed-loop absorption set up had a higher mass transfer coefficient value than the open-loop counterpart, which is an expected behavior as the closed-loop recirculates the water, making it more saturated with , whereas the open-loop set up continuously supplies fresh water.

## 2. Introduction

The main goals of this lab are to investigate the effects that gas and water flow rates have on the column's pressure drop, as well as to determine the mass transfer coefficients of in various operating conditions. Gas absorption-related experiments are important for engineers to understand, as technology is widely used in many industries. More specifically, gas absorption is used in the coal industry, where a wet scrubber is used to absorb the gas produced by the exhaust gas of the furnace [1]. For this lab, the UOP7 column was used to carry out the experiments, a Hempl apparatus was used to measure the void fraction within the column, a compressed air regulator/valve was used to control the flow rate of air going into the column, an air flow meter was used to control the flow of compressed air going into the column, a flow meter was used to control the flow of going into the column, and a water flow meter was used to control the flow of water going into the column. In addition, a DI water tank was also used to hold the DI water reservoir for the setup, a pump was used to allow water flow into the column, two drain valves were used to drain water from the tank and column, respectively, and a manometer was used to read the pressure difference within the column. The experiment began by characterizing the packed bed. To do this, a graduated cylinder was first filled with a known volume of packing material, and water was slowly added until the empty spaces were filled. The additional water volume was then recorded and used to calculate the void fraction. Then, to determine the pressure drop, all drain valves were closed, and the DI reservoir tank was filled to about three-fourths full with DI water. Air was then introduced at 20 L/min and allowed to flow for 5 minutes to dry the column, and the pressure drop across the column was measured at different gas flow rates using the

manometer. After that, the pump was turned on and valves C1 and C4 were opened to allow liquid flow. The water flow was set to 1.5 L/min, and its level was adjusted to stay above the sampling port and below the air inlet. Different air flow rates and their corresponding pressure drop values were recorded until flooding occurred within the column. This procedure was then repeated for higher water flow rates of 3, 6, 8 L/min. In the second part, absorption and desorption were studied in a closed system. The reservoir was first refilled with DI water, the temperature recorded, and an initial sample was titrated using 0.1N NaOH to determine the starting concentration. The liquid flow was set to 6 L/min, air flow was set to 20 L/min, and was added at a flow rate of 5 L/min. Water samples were then collected over time until the concentration reached a somewhat constant value, and the outlet gas composition was measured with the Hempl apparatus. For desorption, the was turned off, the liquid flow was reduced to 1.5 L/min, and water samples were collected at short time intervals and analyzed by titration. In the third part, absorption was studied in an open system. The reservoir was filled again, and the initial dissolved concentration was measured. The liquid flow was set to 1.5 L/min, the drain valve was opened for continuous operation, and the liquid level was controlled with the drain valve. The air flow was then set to 20 L/min, and was introduced at 5 L/min with the heater block on. Water samples were collected every minute until the stays relatively constant, and the concentration was measured using the Hempl apparatus.

### 3. **Theory**

A packed-bed gas absorber is designed to bring a gas stream into intimate contact with a liquid stream over solid packing, which increases interfacial area for mass transfer. In this experiment, an air/ mixture flows counter-currently to water through a bed of 1cm Raschig rings, thus the column behaviour depends on both the hydrodynamics of flow through the packing and the transfer of across the gas-liquid interface [2]. The packing structure is commonly described by its void fraction, which is measured experimentally as: [2].

As the gas flowrate is increased at a fixed liquid flowrate, the pressure drop across the bed increases because the moving phases experience a resistance within the void space. For single-phase packed beds, a common starting point for pressure-drop analysis is the Ergun relation: [3]. In the lab, the wet bed system is not perfectly represented by a single-phase model, but the same idea remains useful: increasing gas and liquid rates increases the resistance to flow. The lab manual states that when a pressure drop is plotted against gas flowrate at a fixed flow, a sharp break in the slop of the graph marks the flooding point, where gas flow is high enough to hold up liquid inside the bed. The manual also states that when operating near 50% of flooding a better gas-liquid contact is achieved without unstable operation [2].

The absorption of carbon dioxide into water is a mass-transfer process driven by departure from the gas-liquid equilibrium. The transfer rate in the simplified form is: [2].

At equilibrium, the gas side and liquid side compositions are related by Henry's law. NIST tabulates Henry's law data for in water and shows that the equilibrium constant is temperature dependent, therefore solubility changes with operating temperature [4].

The overall transfer process is commonly interpreted with the two-film model, in which must diffuse through a gas film, cross the interface, and then diffuse through a liquid film before reaching the bulk liquid. In resistance form, the overall coefficient is treated as the sum of gas side and liquid side resistances. A U.S. Geological Survey treatment of the two-film model gives the overall liquid side form as: , showing that the reciprocal overall coefficient is the sum of individual film resistances [5]. This is important as it explains why one phase can dominate the overall rate. The same source notes that for absorption into water, the liquid-film resistance can dominate strongly, which is why the measured overall coefficient in this lab is often interpreted mainly in terms of liquid-side mass transfer [5].

The experiment connects this theory to measurable quantities in both phases. The dissolved concentration in the liquid is found by titration with standard NaOH, using [2]. The outlet gas composition is measured volumetrically with the Hempl apparatus [2]. Together, the pressure drop data, titration data and gas analysis data allow the packed bed to be characterized in terms of porosity, flooding behaviour, absorption/desorption trends and the overall mass-transfer performance [2].

Table 2: Variables and their Descriptions

| Variable / Quantity | Typical Units                                                                    | Description                                                                                                            |
|---------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| $\varepsilon$       | dimensionless                                                                    | Void fraction of the packed bed; fraction of the total bed volume that is empty space.                                 |
|                     | $\text{m}^3$                                                                     | Volume of the empty space within the packed bed.                                                                       |
|                     | $\text{m}^3$                                                                     | Total volume occupied by the packed bed section.                                                                       |
| $\Delta P$          | Pa                                                                               | Pressure drop across the packed section of the absorber.                                                               |
| $L$                 | m                                                                                | Length or height of the packed bed over which pressure drop is measured.                                               |
| $\Delta P/L$        | Pa/m                                                                             | Pressure drop per unit packed-bed length.                                                                              |
| $\mu$               | $\text{Pa}\cdot\text{s}$                                                         | Dynamic viscosity of the flowing fluid.                                                                                |
| $U$                 | m/s                                                                              | Superficial velocity of the fluid through the column.                                                                  |
|                     | m                                                                                | Characteristic particle or packing diameter; here related to the Raschig-ring size.                                    |
| $\rho$              | $\text{kg}/\text{m}^3$                                                           | Fluid density.                                                                                                         |
|                     | $\text{mol}/(\text{m}^2\cdot\text{s})$ or $\text{mol}/(\text{m}^3\cdot\text{s})$ | Rate of $\text{CO}_2$ transfer; exact unit depends on whether it is written per interfacial area or per packed volume. |

|               |                                                       |                                                                                                                                       |
|---------------|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| k             | m/s or coefficient-dependent                          | Mass-transfer coefficient in the simplified rate expression; exact units depend on the driving-force form.                            |
| a             | $\text{m}^2/\text{m}^3 = 1/\text{m}$                  | Interfacial area per unit packed volume available for gas-liquid mass transfer.                                                       |
| ka            | 1/s                                                   | Volumetric mass-transfer coefficient; combines transfer rate and contact area effects.                                                |
| $\Delta C$    | $\text{mol}/\text{m}^3$                               | Concentration driving force for $\text{CO}_2$ transfer between gas and liquid phases.                                                 |
|               | m/s                                                   | Overall liquid-side mass-transfer coefficient.                                                                                        |
|               | m/s                                                   | Liquid-film mass-transfer coefficient.                                                                                                |
|               | varies with gas driving-force basis                   | Gas-film mass-transfer coefficient; exact units depend on whether the gas driving force is written in pressure or concentration form. |
| R             | $\text{J}/(\text{mol} \cdot \text{K})$                | Universal gas constant.                                                                                                               |
| T             | K                                                     | Absolute temperature.                                                                                                                 |
| H             | $\text{Pa} \cdot \text{m}^3/\text{mol}$ or equivalent | Henry's law constant for $\text{CO}_2$ in water; exact units depend on the Henry's-law convention used.                               |
|               | $\text{mol}/\text{L}$ or $\text{mol}/\text{m}^3$      | Dissolved $\text{CO}_2$ concentration determined from titration.                                                                      |
|               | L or mL                                               | Volume of titrant (standard NaOH) used in the titration.                                                                              |
|               | $\text{mol}/\text{L}$                                 | Concentration of the titrant, usually standard NaOH.                                                                                  |
|               | L or mL                                               | Volume of the liquid sample analyzed in the titration.                                                                                |
| $\text{CO}_2$ | —                                                     | Carbon dioxide, the solute being absorbed into water.                                                                                 |
| NaOH          | $\text{mol}/\text{L}$ when used as concentration      | Sodium hydroxide titrant used to determine dissolved $\text{CO}_2$ content.                                                           |

#### 4. Results & Data Analysis

For the first experiment a series of measurements were taken for pressure drop as a function of both water and gas flowrates. The void fraction was obtained by filling a graduated cylinder with a given value of Raschig rings and adding water until the Raschig rings were covered. The void fraction of the rings was then calculated by dividing the amount of water added to the total volume.

|                      |            |
|----------------------|------------|
| <b>Void fraction</b> | <b>46%</b> |
|----------------------|------------|

Following the determination of the void fraction the pressure drop across the column was measured while the bed was dry. As shown in the figure below, the pressure drop increased as the gas flowrate increased. This is the expected behavior because as gas flowrate increases through the packed regions it creates greater resistance.

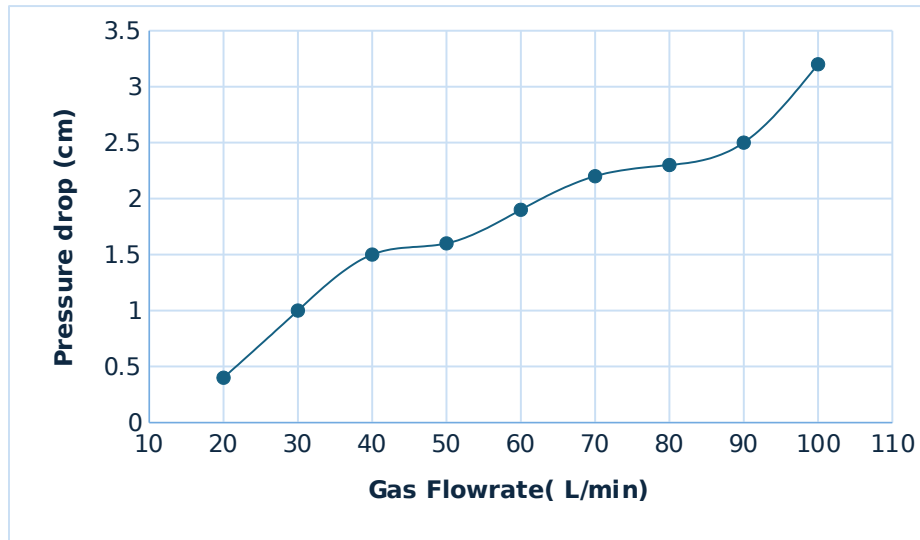


Figure 1 Pressure Drop vs Gas Flowrate for no water

Various flowrates of the liquid water were tested alongside the gas flowrates to determine the effects on the pressure drop and flooding of the column. As shown in fig. 2 increasing the water flowrate caused the pressure drop in the column to rise more quickly and cause flooding to occur at lower gas flowrates. The 1.5 L/min run flooded at the highest gas flowrate while resulting experiments flooded at corresponding lower flowrates.

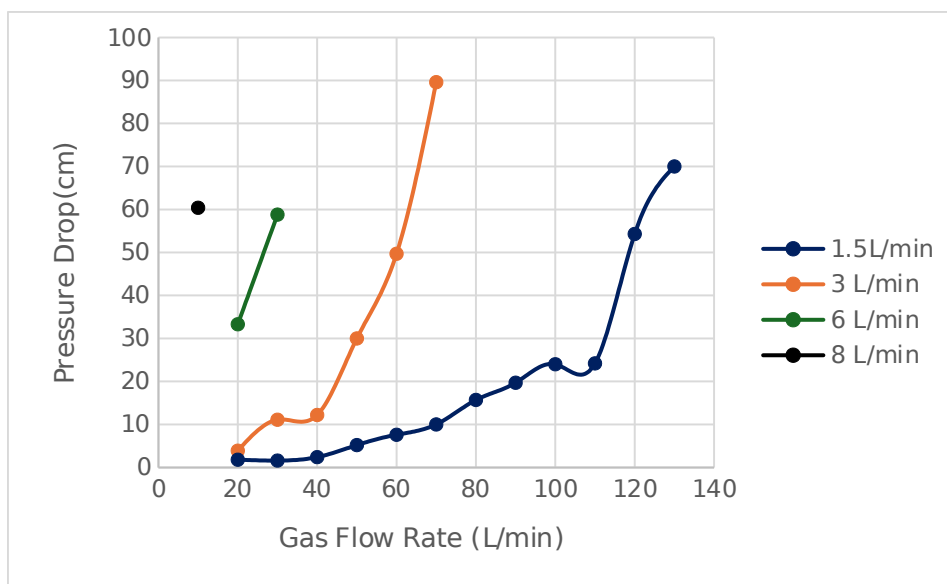


Figure 2: Pressure Drop vs Gas Flow rates for varying water flow rates

For the second experiment absorption of CO<sub>2</sub> in water, the compositions of the dissolved CO<sub>2</sub> in water was determined via titration until a steady state was reached. As shown figure 3 below the concentration of dissolved co<sub>2</sub> initially increased and then varied across a smaller range.

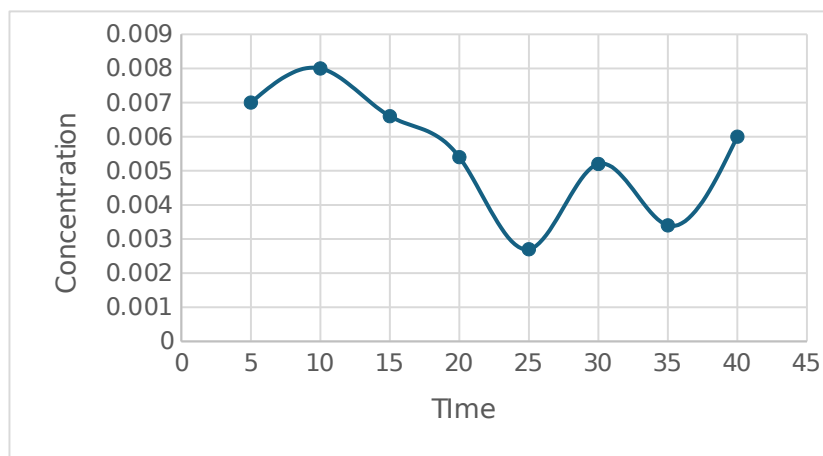


Figure 3: CO<sub>2</sub> Concentration vs Time for EXP 2A

The graphs for experiments 2b and 3 are also pictured below in figures 4 and 5:

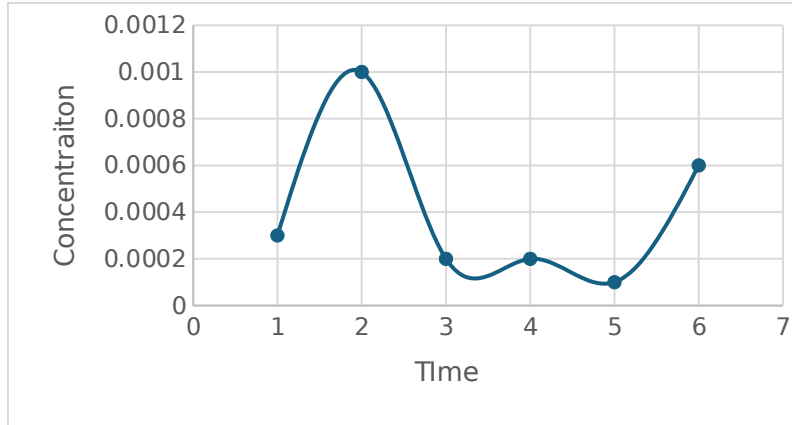


Figure 4 Concentration vs time for EXP 2b

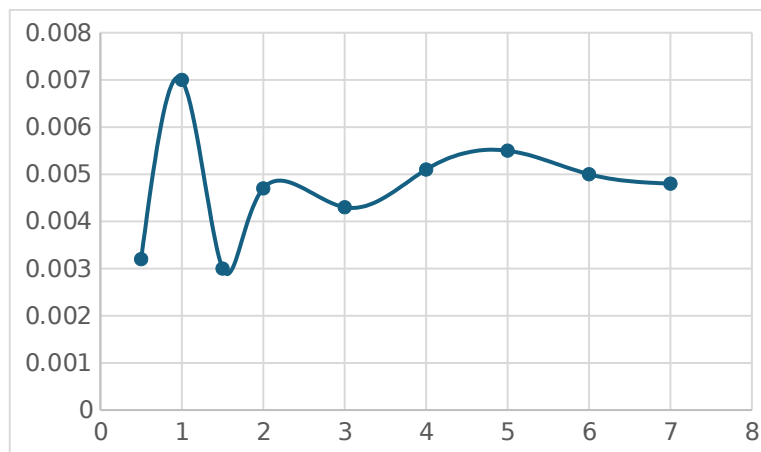


Figure 5 Concentration vs time for EXP 3

From Table 1, the calculated mass transfer coefficients were used to compare the magnitude of the CO<sub>2</sub> transfer in each section. Using the simplified class correlation in 2a gave a coefficient of 0.3674, 2b with 0.861, and 3 had a transfer coefficient of 0.00535. Since a large coefficient corresponds to a faster apparent transfer of mass these results show that 2b had the strongest absorption/desorption.

The 2a value suggests that the closed loop configuration had a relatively strong transfer of mass where in part 3 the apparent absorption was much lower in the open loop configuration. This shows that the closed loop configuration promoted higher overall gas liquid transfer than the open loop configuration in the experiment. The highest value was obtained in 2b under desorption, which is logical due to the high concentration of CO<sub>2</sub> absorbed in the water at the start of the trial, corresponding to a larger driving force.

## 5. Discussion

The packed bed results show that pressure drop increases with gas flowrate, both in the dry bed and with liquid present, this is consistent with the theory due to the

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increased frictional resistance and inertial effects. The measured void fraction 46% is in the range for Raschig rings, this shows a reasonable packing behavior. When water was introduced the pressure drop increased more rapidly and flooding occurred at lower gas flow rates as shown in figure 2. This is expected as liquid holdup reduces available void space and increases resistance to gas flow. The sharp rise in pressure drop at higher gas flow rates indicates the transition from loading to flooding. These trends support standard design practices

The CO<sub>2</sub> absorption data shows that liquid-phase concentration increased at first and approached a steady state. This lines with mass transfer theory as the driving force decreases near equilibrium. The desorption case in part 2B had the highest mass transfer coefficient of 0.861 followed by the closed loop absorption at 0.3674, while the open loop system was 0.00535. The differences can be explained by variations in the driving force. The desorption system began with a high CO<sub>2</sub> concentration in the liquid which meant a more rapid transfer, while the open loop system was constantly introducing fresh water which limited the CO<sub>2</sub> buildup. This is consistent with the relation where transfer rate depends on the concentration gradient.

The trends agree with the theory, there are some discrepancies that arose from the experiment. Fluctuations in flowrates, inaccuracies in titrations, and assumptions such as ideal mixing may have affected the results. Despite the limitations, the experiment demonstrates that both pressure drops, flooding, and mass transfer are strongly influenced by operating conditions. This reinforces principles used in the design and operation of packed bed absorbers.

## 6. **References**

- [1] Sorrels, J. L. (2021, April). *Wet and dry scrubbers for acid gas control* (Section 5, Chapter 1). U.S. Environmental Protection Agency, Office of Air Quality Planning and Standards.
- [2] *Gas Absorption in a Packed Bed Lab Manual*, ChE 382 Unit Operations Laboratory, Spring 2024.
- [3] Motil, B. *Packed Bed Reactor Experiment*; NASA Technical Reports Server, 2019. Used for the Ergun packed-bed pressure-drop relation and viscous/inertial interpretation. ([NASA Technical Reports Server](#))

[4] NIST Chemistry WebBook, *Carbon Dioxide: Henry's Law Data*. Used for the equilibrium relationship between gas-phase CO<sub>2</sub> and dissolved CO<sub>2</sub> in water, and for the temperature dependence of CO<sub>2</sub> solubility. ([NIST WebBook](#))

[5] Rathbun, R. E. *Application of the Two-Film Model to the Volatilization of Acetone and t-Butyl Alcohol from Water as a Function of Temperature*; U.S. Geological Survey Water-Supply Paper 2318. Used for the resistance-in-series form of the two-film model and discussion of dominant liquid-film resistance.

## 7. Appendix I: Data Tabulations/Graphs

| No water             |                   |                   |  |  |  |  |  |  |  |  |  |  |  |  |
|----------------------|-------------------|-------------------|--|--|--|--|--|--|--|--|--|--|--|--|
| Gas flow rate, L/min | Pressure drop, mm | Pressure drop, cm |  |  |  |  |  |  |  |  |  |  |  |  |
| 20                   | 2                 | 0.4               |  |  |  |  |  |  |  |  |  |  |  |  |
| 30                   | 5                 | 1                 |  |  |  |  |  |  |  |  |  |  |  |  |
| 40                   | 7.5               | 1.5               |  |  |  |  |  |  |  |  |  |  |  |  |
| 50                   | 8                 | 1.6               |  |  |  |  |  |  |  |  |  |  |  |  |
| 60                   | 9.5               | 1.9               |  |  |  |  |  |  |  |  |  |  |  |  |
| 70                   | 11                | 2.2               |  |  |  |  |  |  |  |  |  |  |  |  |
| 80                   | 11.5              | 2.3               |  |  |  |  |  |  |  |  |  |  |  |  |
| 90                   | 12.5              | 2.5               |  |  |  |  |  |  |  |  |  |  |  |  |
| 100                  | 16                | 3.2               |  |  |  |  |  |  |  |  |  |  |  |  |

| With water           |                   | 1.5 L water/min |                      |                   | 3L water/min |                      |                   | 6L water/min |                      |                   | 8L water/min |
|----------------------|-------------------|-----------------|----------------------|-------------------|--------------|----------------------|-------------------|--------------|----------------------|-------------------|--------------|
| Gas flow rate, L/min | Pressure drop, cm |                 | Gas flow rate, L/min | Pressure drop, cm |              | Gas flow rate, L/min | Pressure drop, cm |              | Gas flow rate, L/min | Pressure drop, cm |              |
| 20                   | 1.8               |                 | 20                   | 3.9               |              | 20                   | 33.3              |              | 10                   | 60.4              | FLOODED      |
| 30                   | 1.6               |                 | 30                   | 11.1              |              | 30                   | 58.8              | FLOODED      |                      |                   |              |
| 40                   | 2.4               |                 | 40                   | 12.2              |              |                      |                   |              |                      |                   |              |
| 50                   | 5.2               |                 | 50                   | 30                |              |                      |                   |              |                      |                   |              |
| 60                   | 7.6               |                 | 60                   | 49.7              |              |                      |                   |              |                      |                   |              |
| 70                   | 10                |                 | 70                   | 89.6              | FLOODED      |                      |                   |              |                      |                   |              |
| 80                   | 15.7              |                 |                      |                   |              |                      |                   |              |                      |                   |              |
| 90                   | 19.7              |                 |                      |                   |              |                      |                   |              |                      |                   |              |
| 100                  | 24                |                 |                      |                   |              |                      |                   |              |                      |                   |              |
| 110                  | 24.2              |                 |                      |                   |              |                      |                   |              |                      |                   |              |
| 120                  | 54.3              |                 |                      |                   |              |                      |                   |              |                      |                   |              |
| 130                  | 70                | FLOODED         |                      |                   |              |                      |                   |              |                      |                   |              |

Figure 6: Raw Data of Exp 2a

| Time (min) | Sample # | Sample amount, mL | 0.1 M NaOH needed, mL | Free CO <sub>2</sub> amount |
|------------|----------|-------------------|-----------------------|-----------------------------|
| 5          | 1        | 50                | 3.5                   | 0.007                       |
| 10         | 2        | 50                | 4                     | 0.008                       |
| 15         | 3        | 50                | 3.3                   | 0.0066                      |
| 20         | 4        | 50                | 2.7                   | 0.0054                      |
| 25         | 5        | 50                | 1.35                  | 0.0027                      |
| 30         | 6        | 50                | 2.6                   | 0.0052                      |
| 35         | 7        | 50                | 1.7                   | 0.0034                      |
| 40         | 8        | 50                | 3                     | 0.006                       |

Figure 5: Raw Titration Data of Exp 2a

| Time (min) | Sample # | Sample amount, mL | 0.1 M NaOH needed, mL | Free CO2 amount |
|------------|----------|-------------------|-----------------------|-----------------|
| 1          | 1        | 50                | 0.15                  | 0.0003          |
| 2          | 2        | 50                | 0.5                   | 0.001           |
| 3          | 3        | 50                | 0.1                   | 0.0002          |
| 4          | 4        | 50                | 0.1                   | 0.0002          |
| 5          | 5        | 50                | 0.05                  | 0.0001          |
| 6          | 6        | 50                | 0.3                   | 0.0006          |

Figure 6: Raw Titration Data of Exp 2b

| Time (min) | Sample # | Sample amount, mL | 0.1 M NaOH needed, mL | Free CO2 amount |
|------------|----------|-------------------|-----------------------|-----------------|
| 0.5        | 1        | 50                | 1.6                   | 0.0032          |
| 1          | 2        | 50                | 3.5                   | 0.007           |
| 1.5        | 3        | 50                | 1.5                   | 0.003           |
| 2          | 4        | 50                | 2.35                  | 0.0047          |
| 3          | 5        | 50                | 2.15                  | 0.0043          |
| 4          | 6        | 50                | 2.55                  | 0.0051          |
| 5          | 7        | 50                | 2.75                  | 0.0055          |
| 6          | 8        | 50                | 2.5                   | 0.005           |
| 7          | 9        | 50                | 2.4                   | 0.0048          |

Figure 7: Raw Titration Data of Exp 3

## 8. Appendix II: Error Analysis

Several sources of error likely affected the data collected in the experiment. The pressure drop data depended on reading the manometer while adjusting gas and water flowrates, so small reading error or slight changes in the flow could shift the plotted values and change the actual flooding point. This was especially important near flooding where the pressure drop began increasing very quickly. The same issue applies to the absorption data, as observed the flow rate of all three components drifted slightly as the experiment continued. This is likely the contributing factor in the scattered dissolved CO<sub>2</sub> values from titration.

Titration also introduced uncertainty because the endpoint and final determination of the concentration is based on a slight color variation in the solution, so small differences in amount of titrant used have a exacerbated effect on the solution. This would directly affect the mass transfer coefficients since those values are based on the concentrations measured. In 2b, the likelihood for error was larger because the outlet gas concentration was not measured directly and was instead estimated from the first desorption interval.

A numerical look at the data shows a noticeable scatter in the results obtained. In the dried run the pressure drop increased from 0.4 cm at 20 L/min to 3.2 at 100 L/min but the curve became steeper at higher flows where minute changes in flow or reading would have an exacerbated effect. The absorption data showed even more variations, in 2a the

dissolved CO<sub>2</sub> concentration ranged from 0.0027 to 0.0080 and the calculated coefficients ranges from 0.1581 to 0.5561 to an average of 0.3674. Part three showed similar variations with ranges from 0.00137 to 0.01057 to an average of 0.00535. This shows that the reported mass transfer coefficients are best used to compare trials and not to be used as exact coefficients.

#### 9. **Appendix III: Sample Calculations**

To calculate the void table with the following equation using experiment 1 results:

Using part 2a values for the sample calculations:

The dissolved concentration in the water is calculated using the following formula:

is the volume of NaOH used in the titration. Using the 5-minute samples from part 2a:

Using the 5-minute samples to calculate the mass transfer rate from the liquid side:

The overall mass transfer coefficient is calculated using the following:

The same calculation was repeated for each time point in part 2a. Values were averaged to obtain the reported coefficient:

These sample calculations were then repeated for the rest of the experiment.

#### 10. **Appendix IV: Individual Team Contributions**

| ASSIGNMENT                         | TIME (HOURS) | DESCRIPTION                                  |
|------------------------------------|--------------|----------------------------------------------|
| Operator (Both Lab Days)           | 4            | All Team members                             |
| Pre-Lab Editing                    | 2            | All Team members                             |
| Experimental Objective (Prelab)    | 1            | Miguel Piza                                  |
| Apparatus (prelab)                 | 1            | Khoa Nguyen & Rishabh Patel                  |
| Materials & Supplies (prelab)      | 1            | Khoa Nguyen                                  |
| Procedure (prelab)                 | 1            | Alexis Guillen                               |
| Project Safety Evaluation (prelab) | 1            | Kyle Hankosky                                |
| Final Lab Editing                  | 2            | All Team members                             |
| Executive Summary (final lab)      | 2            | Khoa Nguyen & Rishabh Patel                  |
| Introduction (Final lab)           | 1            | Khoa Nguyen & Rishabh Patel                  |
| Theory (Final Lab)                 | 5            | Rishabh Patel & Khoa Nguyen                  |
| Results (final lab)                | 5            | Kyle Hankosky & Alexis Guillen & Miguel Piza |
| Discussion (final lab)             | 2            | Khoa Nguyen & Miguel Piza                    |
| References (final and/or prelab)   | 1            | Rishabh Patel                                |
| Data Tabulation/Graphs (final)     | 2            | Kyle Hankosky & Alexis Guillen & Miguel Piza |
| Error Analysis (final lab)         | 1            | Kyle Hankosky & Alexis                       |

|                                 |          |                                |
|---------------------------------|----------|--------------------------------|
|                                 |          | Guillen                        |
| Sample Calculations (final Lab) | 2        | Kyle Hankosky & Alexis Guillen |
| PowerPoint Presentation         | 5        | All Team members               |
| Total                           | 39 Hours |                                |